

Ch. 3 — P01 (D) Conceptual Model for a Coffee Kiosk

Problem Bank | Chapter 3 | Type: D | Difficulty:

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A university coffee kiosk operates 7:00–10:00 AM on weekdays. Walk-up customers order at one of two registers, wait for a barista to prepare their drink, collect it at the pickup counter, and leave. There are 2 cash registers and 3 baristas. A customer whose order requires more than one barista action (e.g., two espressos) holds one barista until the order is complete.

The simulation study question is: “*How many baristas are needed to keep mean wait-for-drink time below 3 minutes at the 95th percentile during the 8:00–9:00 peak hour?*”

Tasks

1. **Boundary choice:** Identify what is inside and outside the model boundary. Justify your choice given the study question.
2. **Entity table:** List all entity types with their attributes. Include at least 3 attributes per entity.
3. **Resource table:** List all resources with: name, capacity, service time distribution (state plausible parameters), and discipline.
4. **Event list:** List all distinct events with:
 - The state change each causes
 - Which resources are seized or released
5. **State variables:** List the minimal set of state variables needed to resume the simulation from any snapshot.
6. **Performance measures:** Define at least 4 performance measures that address the study question, with mathematical notation.
7. **Assumptions:** Write at least 6 explicit, quantified assumptions. For each, state what modelling error its violation would introduce.
8. **Pre-simulation bound:** The peak-hour arrival rate is 45 customers/hour. With 3 baristas each serving at rate $\mu = 20$ orders/hour, compute the utilisation ρ . Is the system stable at peak load? At what minimum number of baristas does stability hold?